

E-Commerce Sales & Customer Analytics Analysis Report

Project Overview

The E-Commerce Sales & Customer Analytics Dashboard is a Power BI-based business intelligence project developed to analyze and visualize e-commerce sales, product performance, customer behavior, geographical performance, and order and delivery operations. The analysis helps management and other stakeholders understand business performance, identify high-performing areas, recognize potential issues, and support data-driven decision-making.

Purpose

to transform raw e-commerce data into meaningful business insights that aims to:

- Analyze overall sales and revenue performance.
- Identify high-performing products, categories, and brands.
- Understand customer behavior, segments, and purchasing patterns.
- Analyze sales and order performance across different states and cities.
- Monitor order status, cancellations, returns, payment methods, and delivery performance.
- Provide interactive visualizations and KPIs to help stakeholders identify trends, evaluate performance, and make data-driven business decisions.

Data Summary

The project uses an e-commerce dataset consisting of sales, product, and customer information. The datasets are integrated in Power BI to support analysis of sales performance, product and category performance, customer behavior, geographical performance, and order and delivery operations.

- The Sales dataset contains 250,000 records, representing individual sales/order-product transactions.
- The Products dataset contains 2,000 unique products across different categories and brands.

- The Customers dataset contains approximately 40,000 customer records.
- The data includes multiple categories such as Electronics, Home, Sports, Fashion, Grocery, Beauty, and Books.
- Sales transactions contain information about orders, customers, products, quantity, amount, payment mode, order status, and geographical location.
- Customer data supports analysis by customer segment, age group, gender, state, orders, and spending.
- The dataset supports analysis of delivered, cancelled, returned, shipped, and processing orders, along with delivery performance.

Cleaning Summary

The data cleaning process was performed in Power Query to improve the quality, consistency, and reliability of the e-commerce datasets before loading them into the Power BI data model.

The main cleaning activities included:

- Checked and handled missing/null values in relevant columns.
- Identified duplicate records, particularly considering that multiple product rows can belong to the same Order_ID.
- Corrected and standardized data types for dates, numbers, text, and other fields.
- Standardized categorical values such as category, order status, payment mode, gender, and customer-related fields.
- Checked inconsistent or invalid values in important business fields.
- Validated relationships and key fields such as Customer_ID, Product_ID, and Order_ID.
- Created/processed delivery-related data using Order_Date and Delivery_Date for delivery performance analysis.
- Reviewed cancelled and returned orders to ensure they were appropriately represented in the analysis.
- Removed or adjusted unnecessary/inconsistent data where required to prepare the datasets for reporting.

Data Model Overview

The E-Commerce Sales & Customer Analytics Dashboard uses a star-schema data model in Power BI. The Sales table acts as the central fact table, while Products, Customers, and Dim_Date are used as dimension tables. This structure enables efficient and consistent analysis across sales, products, customers, geographical performance, and order and delivery activities.

Data Model:

- **Sales** – Contains transactional information such as Order ID, Customer ID, Product ID, Order Date, Quantity, pricing, discounts, payment mode, order status, and delivery information.
- **Products** – Contains product-related information including Product ID, Product Name, Category, Brand, prices, discounts, stock, and ratings.
- **Customers** – Contains customer information such as Customer ID, customer details, demographics, location, customer tier, orders, and spending.
- **Dim_Date** – A dedicated date dimension containing Date, Year, Quarter, Month, Month Name, and Year-Month fields for time-based analysis.

Relationships:

- All dimension table have a one-to-many relationship with the Sales table

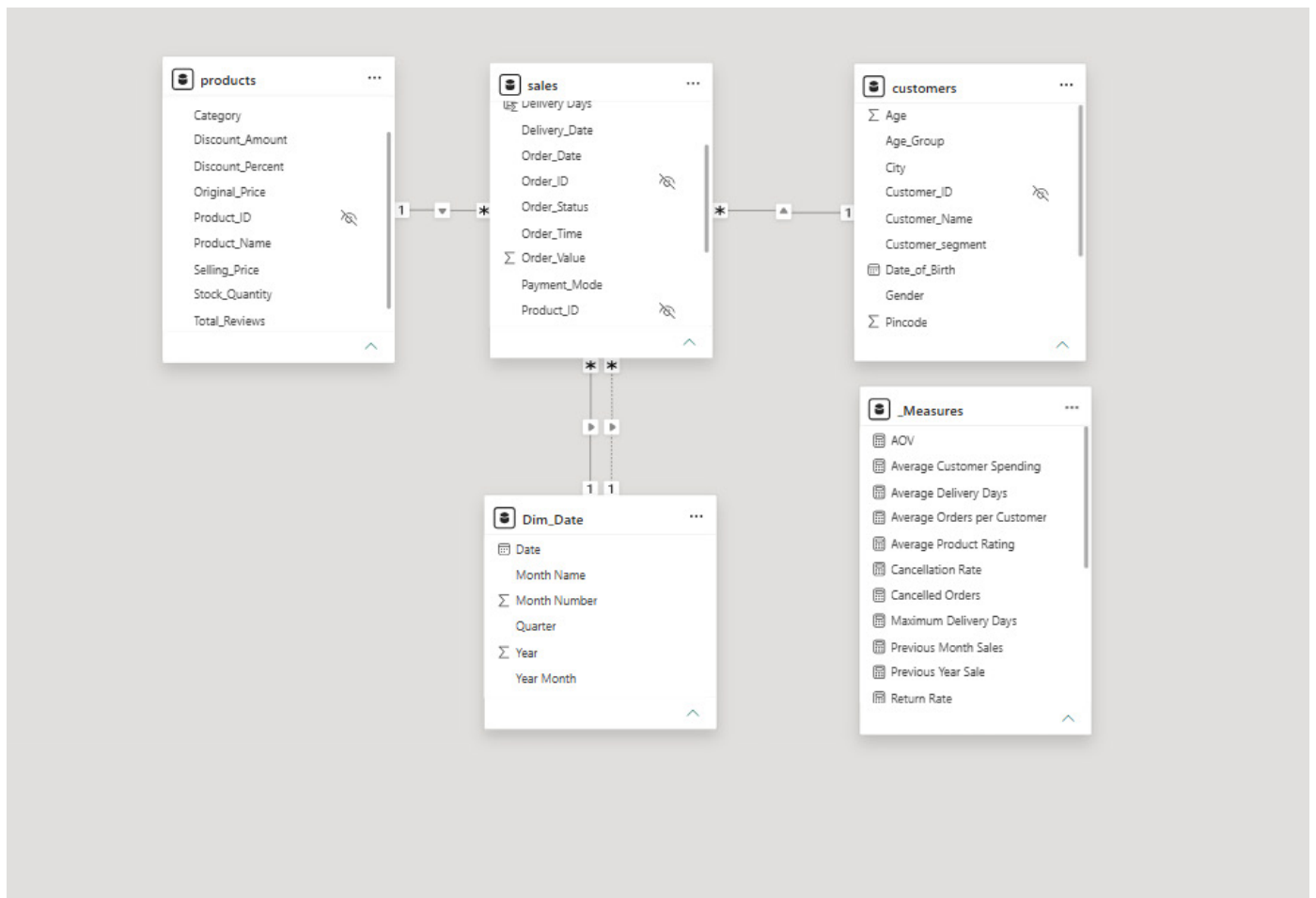
Calculated Measures:

- Total Sales
- Total Orders
- Total Quantity Sold
- Total Customers
- Average Order Value
- Sales Growth %
- Total Discount
- Cancellation Rate
- Return Rate
- Average Delivery Days
- Average Customer Spending
- Average Orders per Customer
- Total Products
- Average Product Rating

Calculated Measures:

- Delivery Days

Model View:



Insights

- **Strong Overall Sales Performance**

The business generated approximately 5.93 billion in total sales from 250,000 orders, indicating a significant overall sales volume.

- **High Customer Engagement**

With approximately 39,914 unique customers and an average of 6.26 orders per customer, customers are making multiple purchases, indicating good customer engagement and repeat purchasing behavior.

- **Customer Spending is Significant**

The average customer spending is approximately 148,586.79, showing that customers contribute substantial revenue over their purchasing activity.

- **Product and Category Performance Varies**

Sales differ across categories, brands, and products. The top-performing products and categories represent important revenue contributors and can be prioritized for inventory and promotional strategies.

- **Customer Segments Show Different Purchasing Behavior**

The dashboard shows sales distribution across Platinum, Gold, and Silver customer segments. This can help the business design targeted offers and loyalty strategies for different customer groups.

- **Geographical Performance is Uneven**

Sales and order volumes vary across states and cities. High-performing locations can be targeted for expansion, while lower-performing regions may require localized marketing strategies.

- **Order Cancellations Require Monitoring**

There were 12,507 cancelled orders, resulting in a cancellation rate of approximately 5%. Reducing cancellations could help improve realized sales and operational efficiency.

- **Returns Have a Business Impact**

The dashboard records 12,493 returned orders, also approximately 5% of total orders. Monitoring return patterns can help identify potential product or customer-service issues.

- **Delivery Performance is Relatively Consistent**

The overall average delivery time is 4.50 days. Delivery performance by state can be monitored to identify locations where logistics improvements may be required.

- **Payment Preferences Can Support Strategy**

The distribution of orders across payment modes provides insight into customer payment preferences and can help the business optimize payment options and checkout processes.

- **Discounting Should Be Evaluated Against Sales**

The discount-versus-sales analysis helps identify whether higher discounts are associated with stronger sales. This can support better promotional and pricing decisions.

- **Discounting Should Be Evaluated Against Sales**

The dashboard reports 4.14% sales growth, indicating positive sales movement compared with the comparison period used in the dashboard.

Conclusion

The E-Commerce Sales & Customer Analytics Dashboard successfully transforms raw e-commerce data into meaningful business insights using Power BI. The project provides a comprehensive view of sales performance, product and category performance, customer behavior, geographical performance, and order and delivery operations.

The use of data cleaning, data modelling, DAX measures, KPIs, and interactive visualizations makes it easier to identify trends, compare performance, and understand key business areas. The dashboard also helps highlight opportunities such as improving customer retention, reducing cancellations and returns, optimizing delivery performance, and focusing on high-performing products and regions.

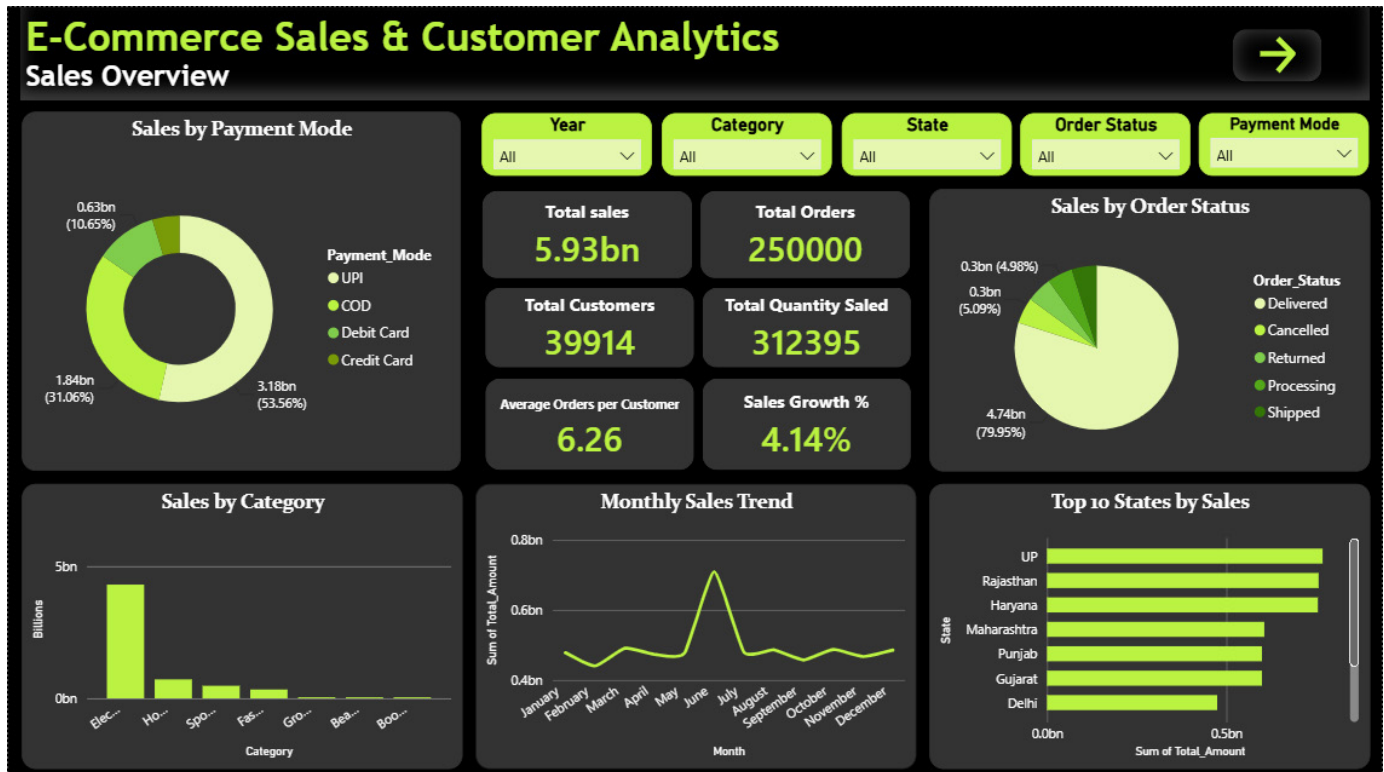
Overall, the project demonstrates how business intelligence and data visualization can support data-driven decision-making and provide a clear understanding of e-commerce business performance.

Recommendations

- Focus on high-performing products to maintain strong sales and availability.
- Improve customer retention through loyalty programs and personalized offers.
- Reduce order cancellations and returns by identifying and addressing common issues.
- Improve delivery performance in locations with longer delivery times.
- Optimize discounts and promotions to increase sales without unnecessarily reducing revenue.
- Focus on high-performing regions and develop strategies to improve low-performing areas.
- Monitor key KPIs regularly to support better and faster business decisions.

Dashboard Overview

Sales Overview :



Purpose:

To provide an overall view of the business's sales performance by analyzing total sales, orders, quantity sold, customers, sales trends, categories, payment modes, and order status.

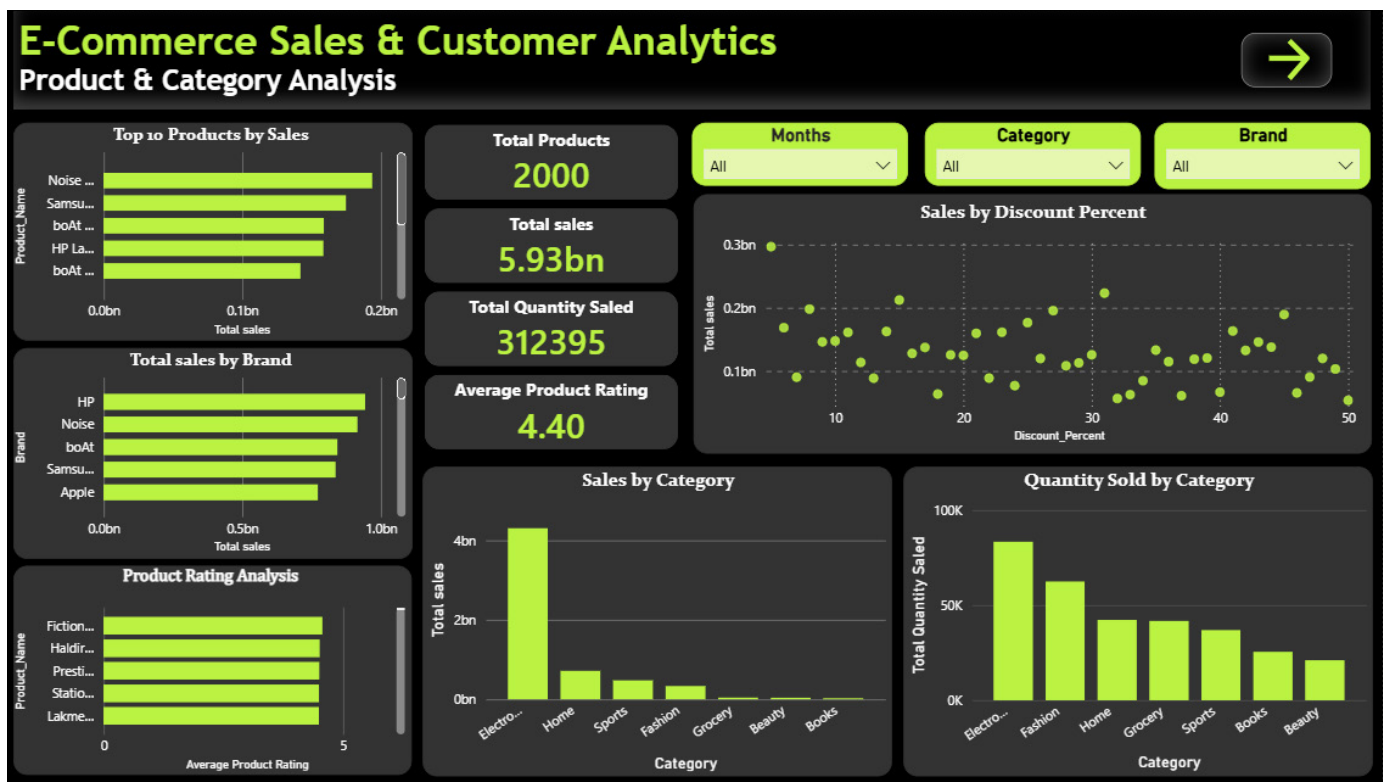
Visuals Used:

- **KPI Cards** - Total Sales, Total Orders, Total Customers, Total Quantity Sold, Average Order per customer, Sales Growth %
- **Line Chart** - Monthly Sales Trend
- **Bar Chart** - Top States by Sales
- **Column Chart** - Sales by Category
- **Pie Chart** - Sales by Order Status
- **Donut Chart** - Sales by Payment Mode
- **Slicers** - Year, Category, State, Order Status, Payment Mode

Insights Provided :

- Provides an overall view of sales and order performance.
- Shows monthly sales trends and sales growth.
- Identifies top-performing states and categories.
- Provides insights into customer order volume and purchasing activity.
- Shows customer payment preferences.
- Highlights the distribution of order statuses.

Product & Category Analysis :



Purpose:

To evaluate the performance of products, categories, and brands by analyzing sales, quantity sold, product ratings, and discounts, helping identify high- and low-performing products.

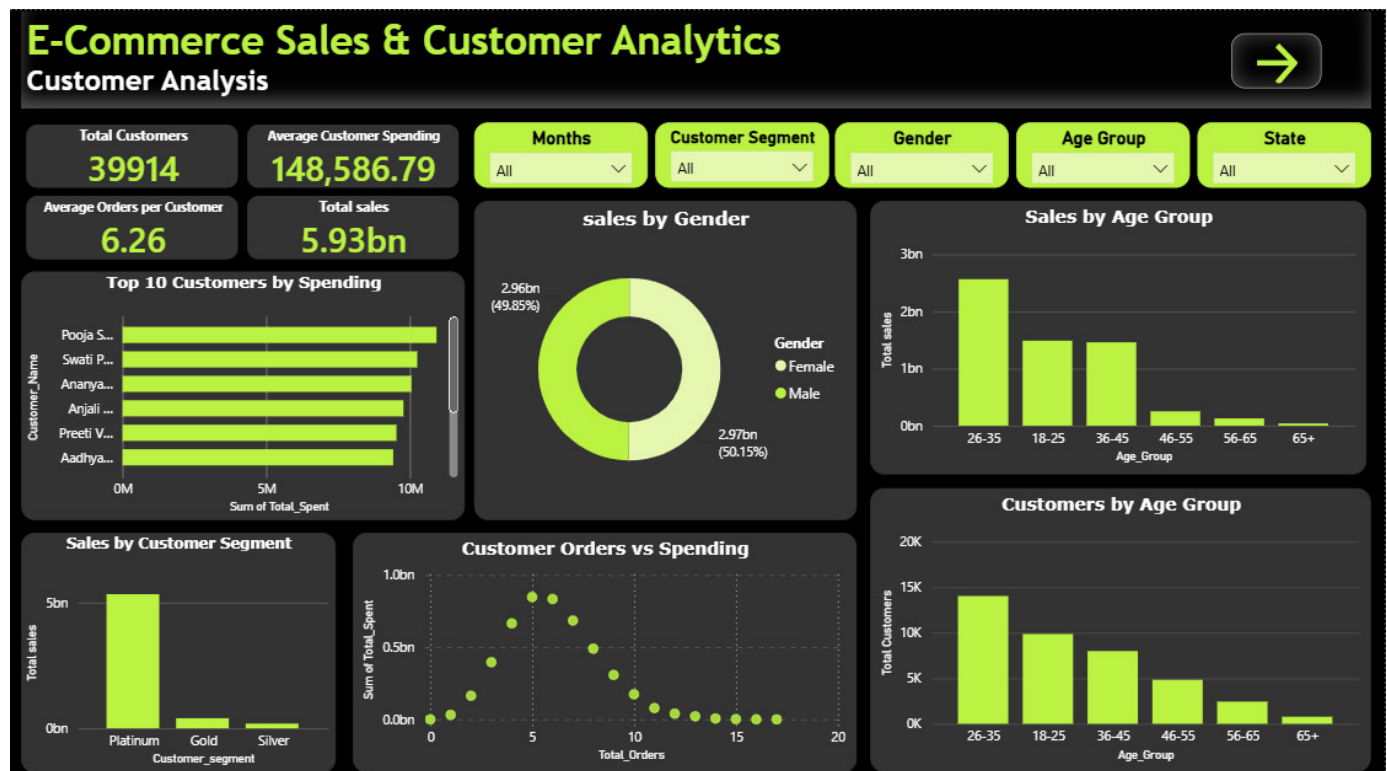
Visuals Used:

- **KPI Cards** - Total Sales, Total Products, Total Quantity Sold, Average Product Rating
- **Scatter Chart** - Sales by Discount Percentage
- **Bar Chart** - Top 10 Products by Sales, Total Sales by Brand, Product Rating Analysis
- **Column Chart** - Sales by Category, Quantity Soled by Category
- **Slicers** - Months, Category, Brand

Insights Provided :

- Identifies top-performing products based on sales.
- Compares sales across categories and brands.
- Shows quantity sold by category.
- Helps evaluate the relationship between discounts and sales.
- Provides insights into product ratings.
- Helps identify products and categories that require greater attention.

Customer Analysis :



Purpose:

To understand customer purchasing behavior by analyzing customer segments, spending, order frequency, age groups, gender, and top customers.

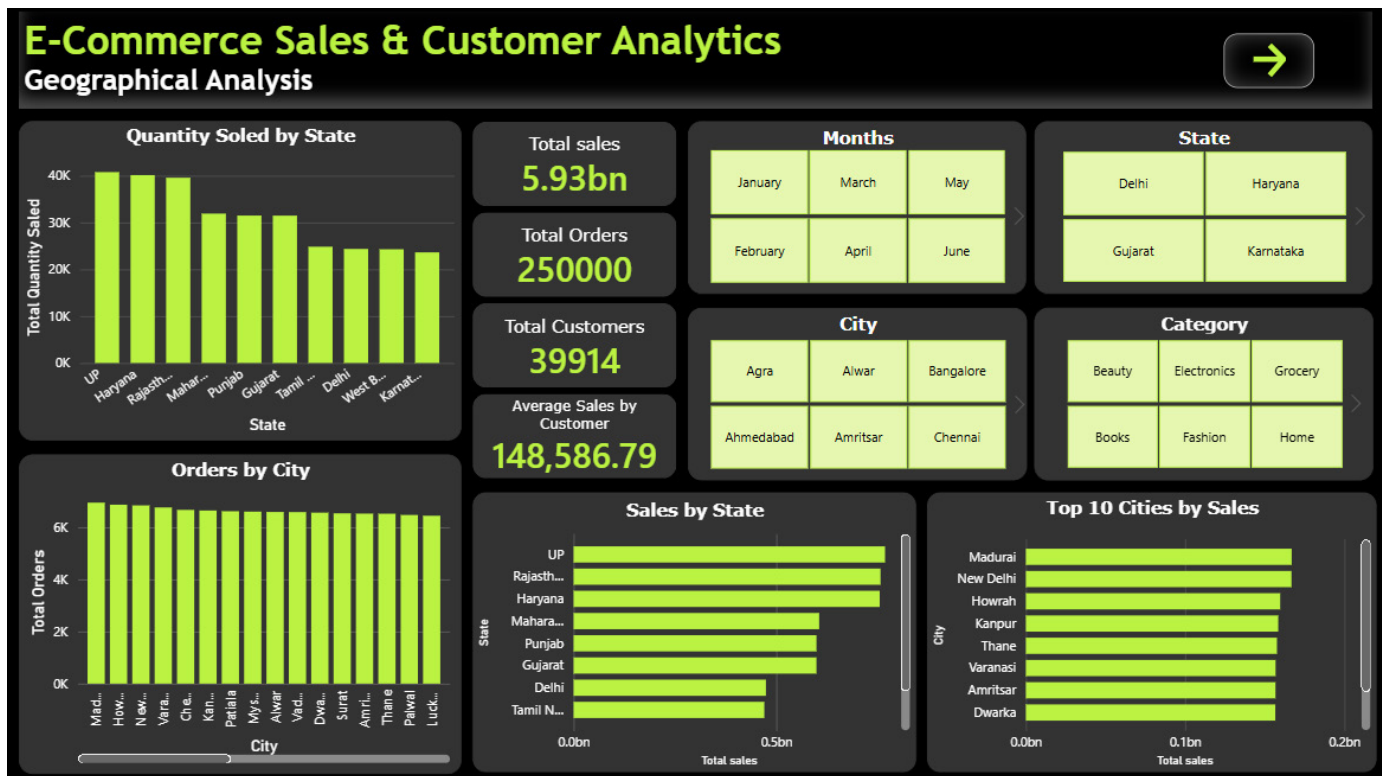
Visuals Used:

- **KPI Cards** - Total Customers, Total Sales, Average Customer Spending, Average Orders per Customer
- **Scatter Chart** - Customer Orders vs Spending
- **Bar Chart** - Top 10 Customers by Spending
- **Column Chart** - Sales by Customer Segment, Sales by Age-group, Customers by Age-group
- **Donut Chart** - Sales by Gender
- **Slicers** - Months, Category, Brand

Insights Provided :

- Shows sales performance across customer segments such as Platinum, Gold, and Silver.
- Identifies top customers by spending.
- Analyzes customer behavior across different age groups and genders.
- Shows the relationship between customer orders and sales/spending.
- Helps understand customer purchasing patterns and valuable customer groups.

Geographical Analysis :



Purpose:

To analyze sales and order performance across different states and cities, helping identify high-performing and low-performing geographical regions.

Visuals Used:

- **KPI Cards** - Total Sales, Total Orders, Total Customers, Average Sales by Customer
- **Bar Chart** - Sales by State, Top 10 Cities by Sales
- **Column Chart** - Quantity Sold by State, Orders by City
- **Slicers** - Months, State, City, Category

Insights Provided :

- Identifies high- and low-performing states based on sales.
- Shows order distribution across cities.
- Compares sales and quantity sold across locations.
- Helps identify geographical areas with stronger customer demand.
- Supports decisions related to regional marketing and business expansion.

Order & Delivery Analysis :



Purpose:

To monitor order and delivery performance by analyzing order status, cancellations, returns, payment methods, and delivery time across different locations.

Visuals Used:

- **KPI Cards** - Total Orders, Cancellation Rate, Return Rate, Average Delivery Days, Total Shipping Cost
- **Column Chart** - Orders by Payment Mode
- **Line Chart** - Orders by Month, Cancelled Orders by Month
- **Donut Chart** - Orders by Status
- **Slicers** - Year, Order Status, Payment Mode, State

Insights Provided :

- Provides insights into order status, including completed, cancelled, and returned orders.
- Shows the cancellation and return rates.
- Identifies payment mode preferences.
- Analyzes average delivery time by state.
- Shows monthly order and cancellation trends.
- Helps identify areas where order processing and delivery performance can be improved.

Limitations

- The project uses only the available sales, product, and customer datasets.
- The dashboard does not provide real-time data.
- Detailed reasons for cancellations are not available.
- Detailed delivery and courier information is not included.
- The project does not include sales forecasting or predictive analysis.
- The accuracy of the results depends on the quality of the source data.
- External factors such as competitors, market trends, and economic conditions are not considered.